

Sketching Activity

DATA 3302: Data Visualization

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Cal Poly San Luis Obispo · Fall 2026

How many ways can you visualize these two numbers:

75

37

*Sketching is a quick, inexpensive, disposable way
of generating, evaluating, and sharing ideas.*

Buxton, Sketching User Experiences, 2007

*Consult examples early, repeated exposure to
examples improves creativity.*

Kulkarni, 2012

*After you know the norms and the rules you must
learn **how to break them***

*How many **more** ways can you visualize these two numbers:*

75

37

If you're stuck, introduce a constraint: one line, only black/white, only round objects, etc.

Some examples

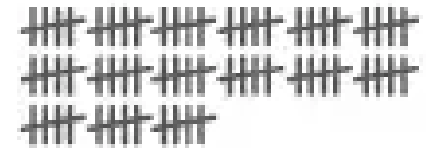
<https://blog.visual.ly/45-ways-to-communicate-two-quantities/>

75, 37

a

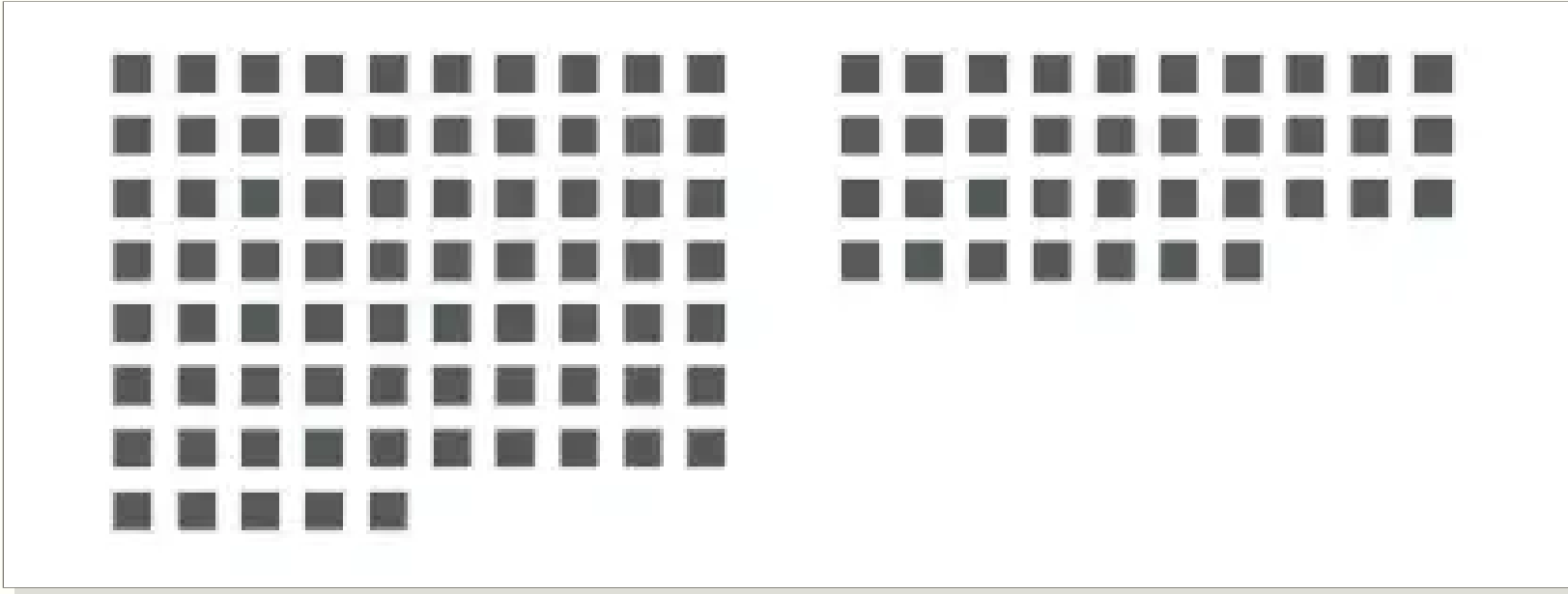


b

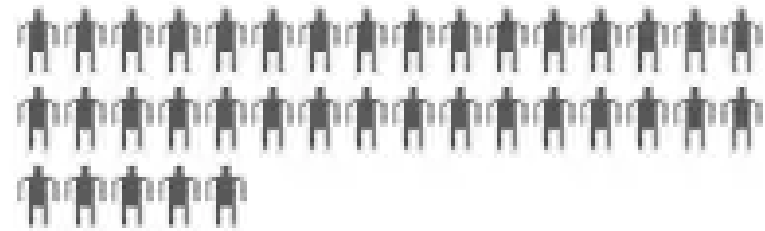
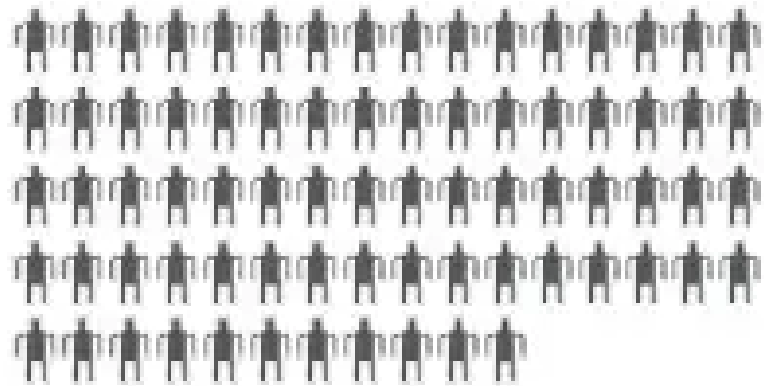


c

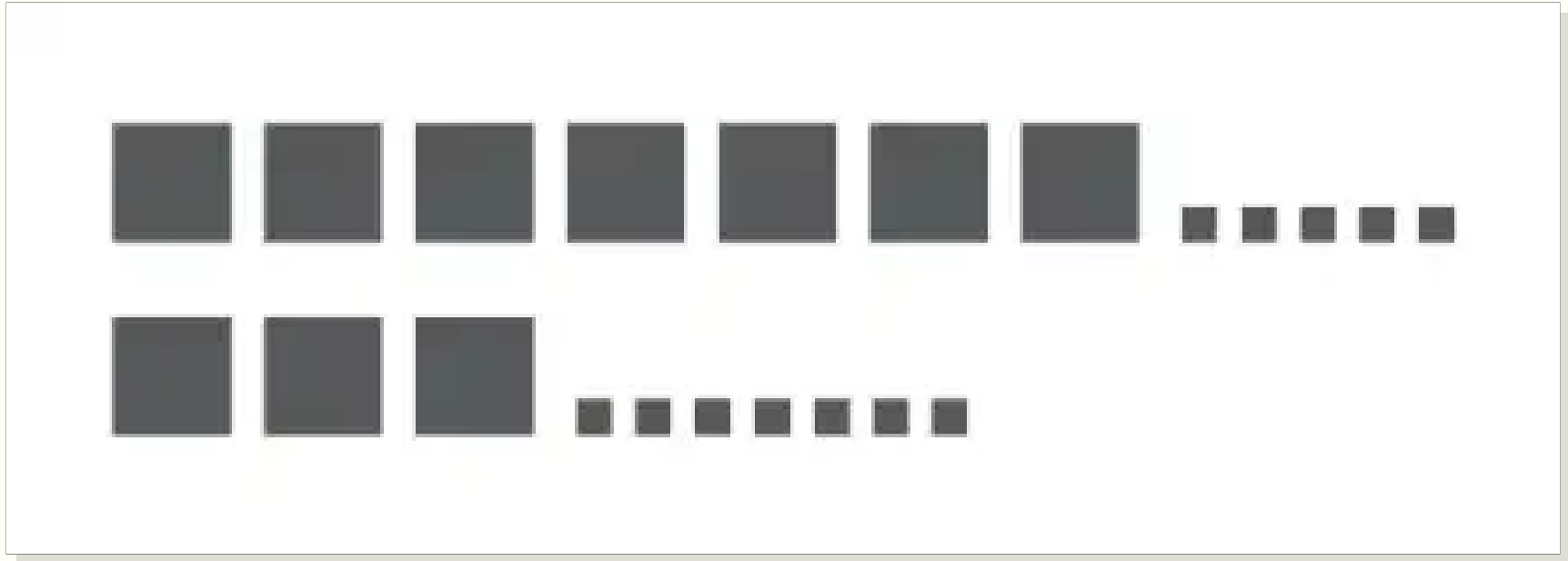
Number notation



Squares



Repeated icon



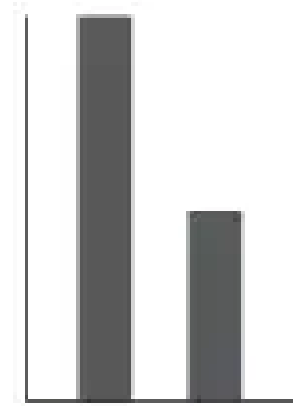
Positional squares



a

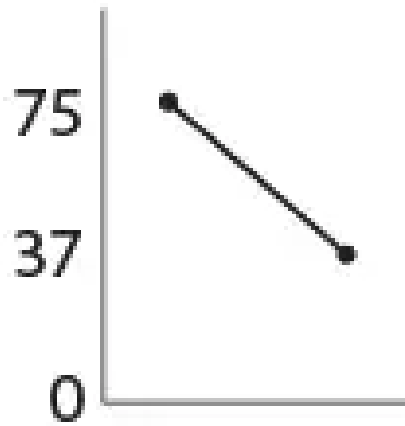


b

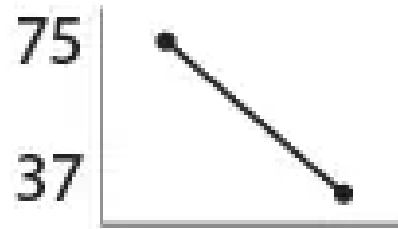


c

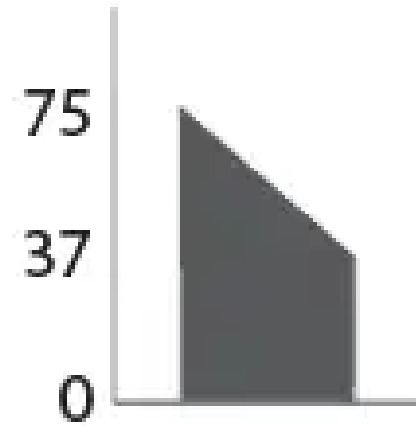
Bars



a



b

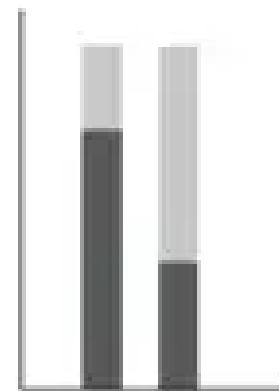


c

Line graph

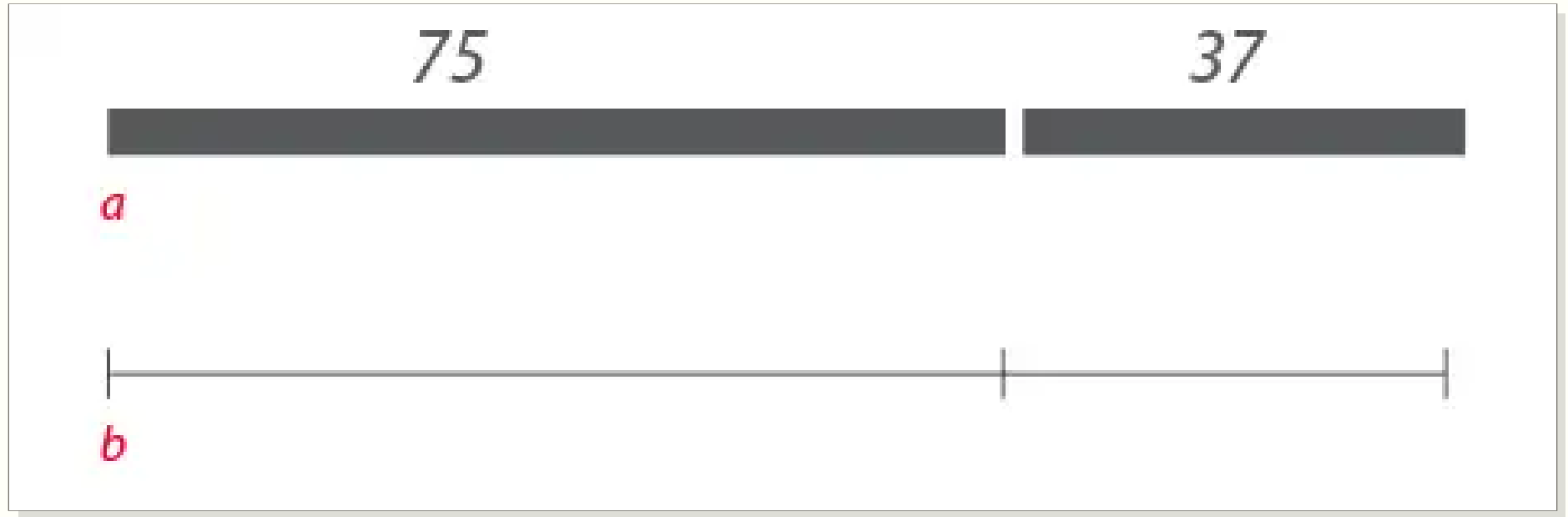


a

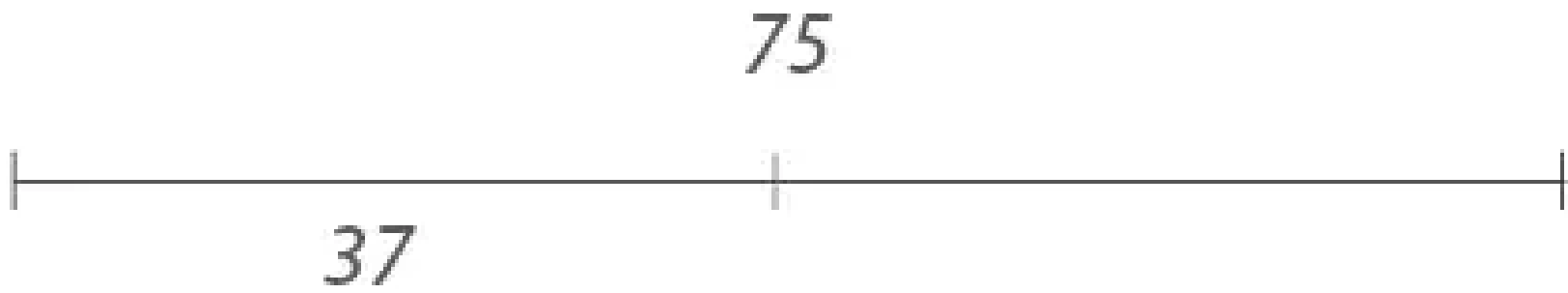


b

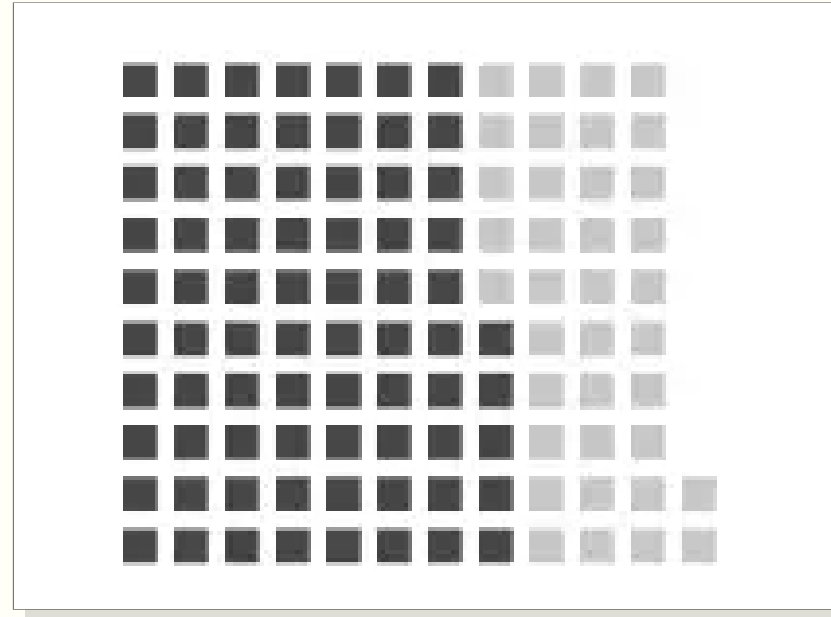
Percentage bars



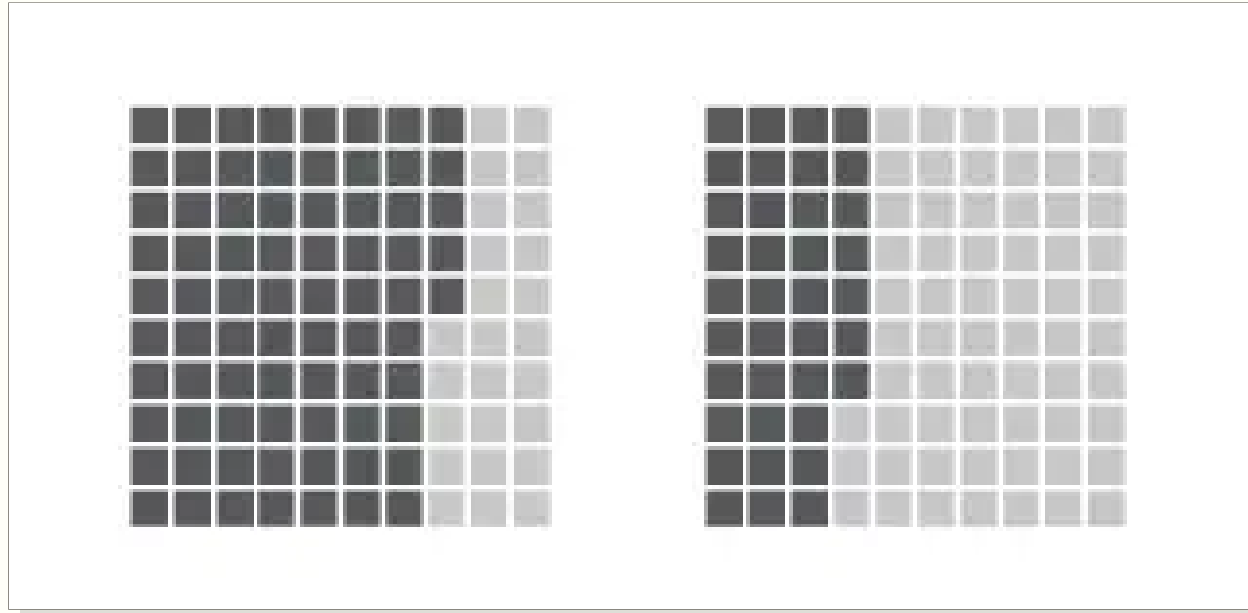
Spliced bar



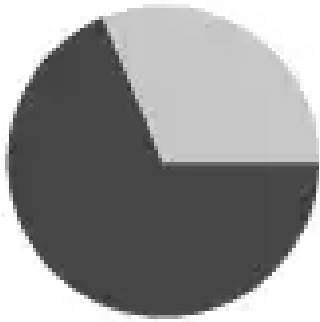
Proportion



Squares merged



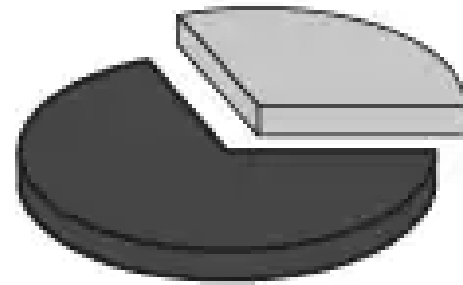
Percentages in squares



a

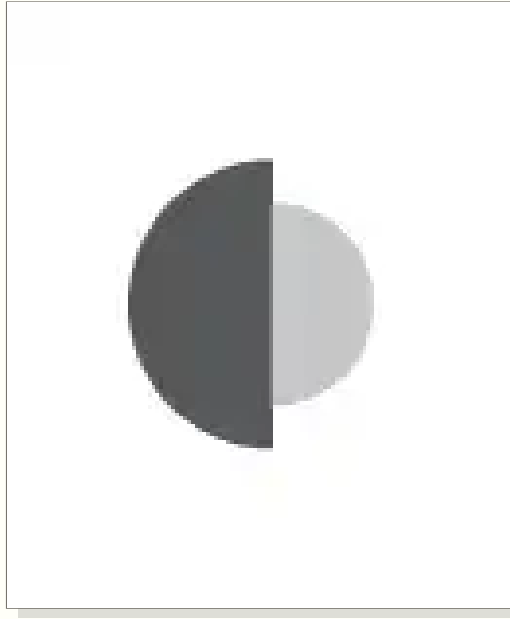


b

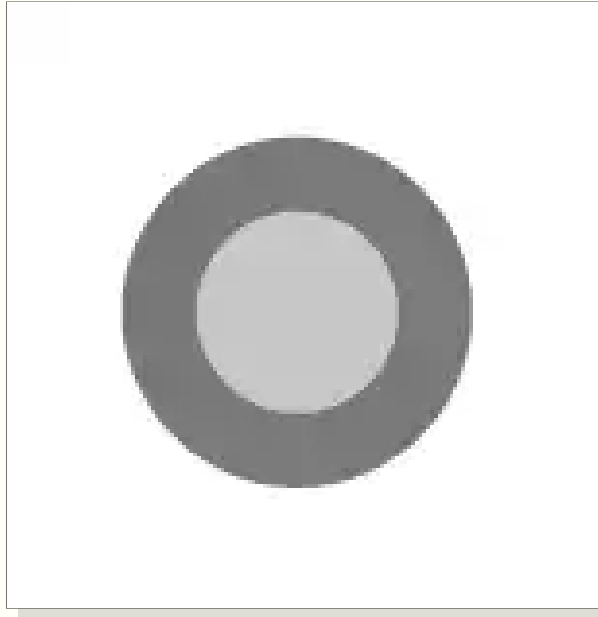


c

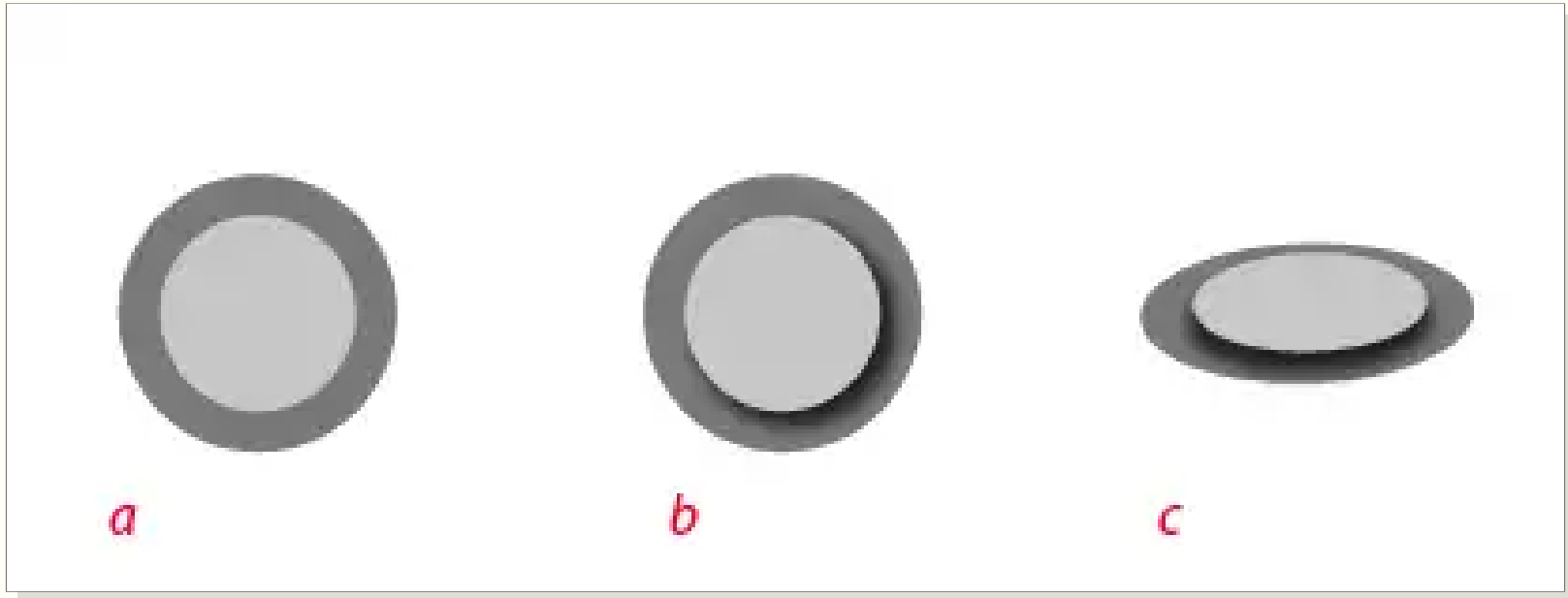
Pie charts



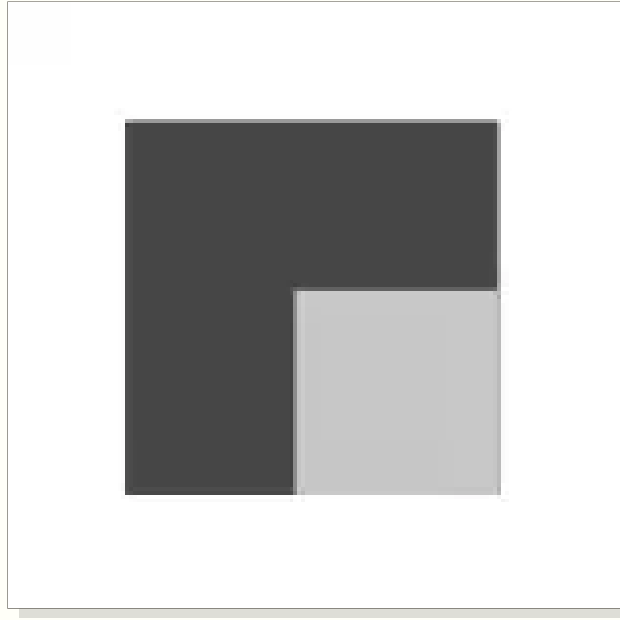
Semi-circle areas



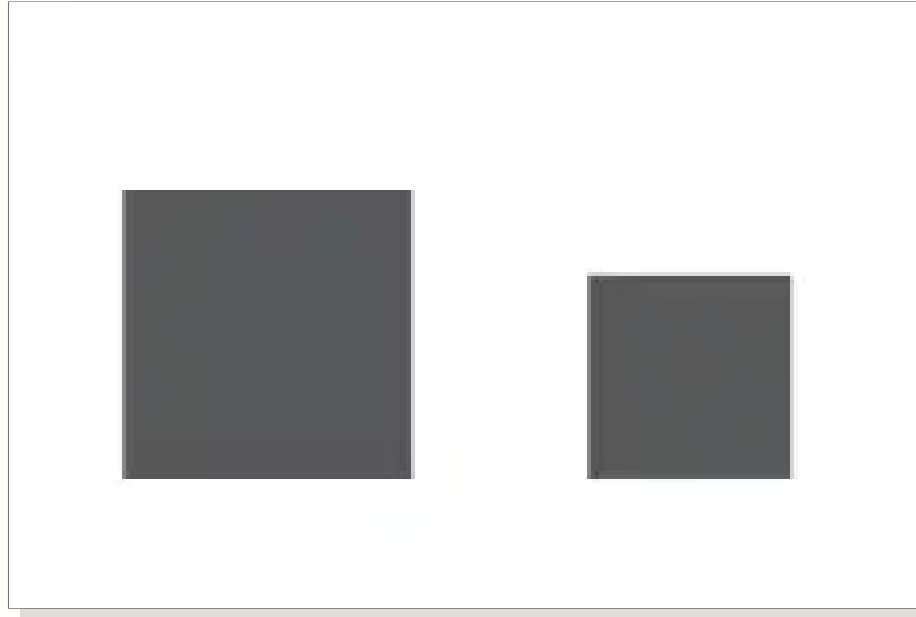
Circle and external ring



Co-centered circles



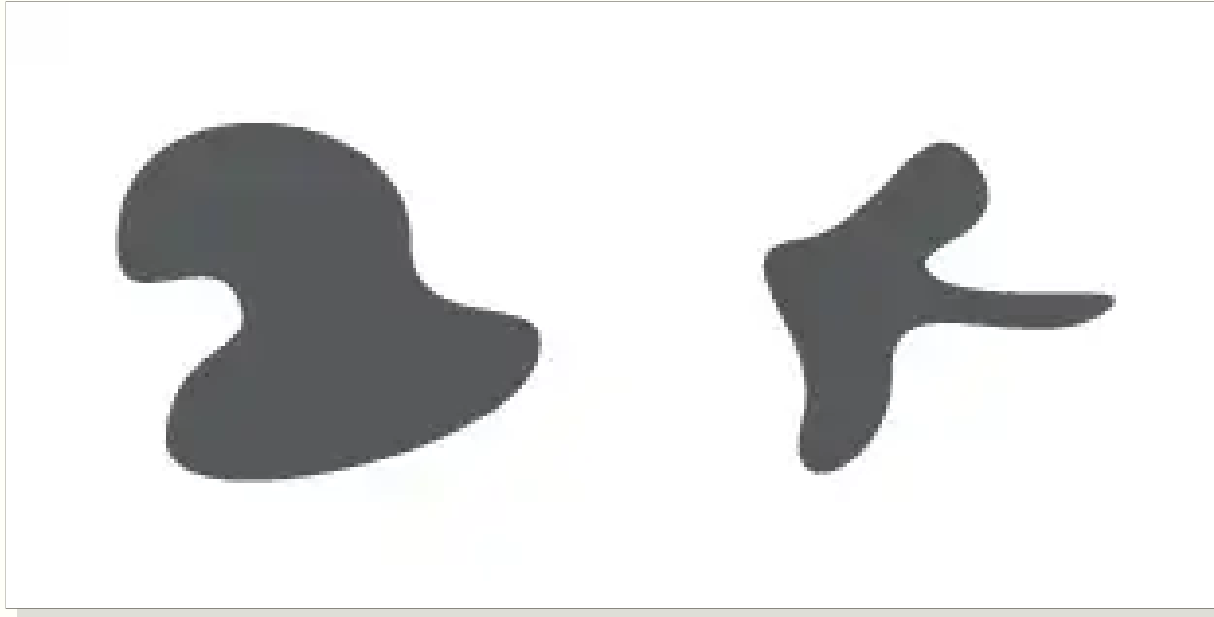
Square divided



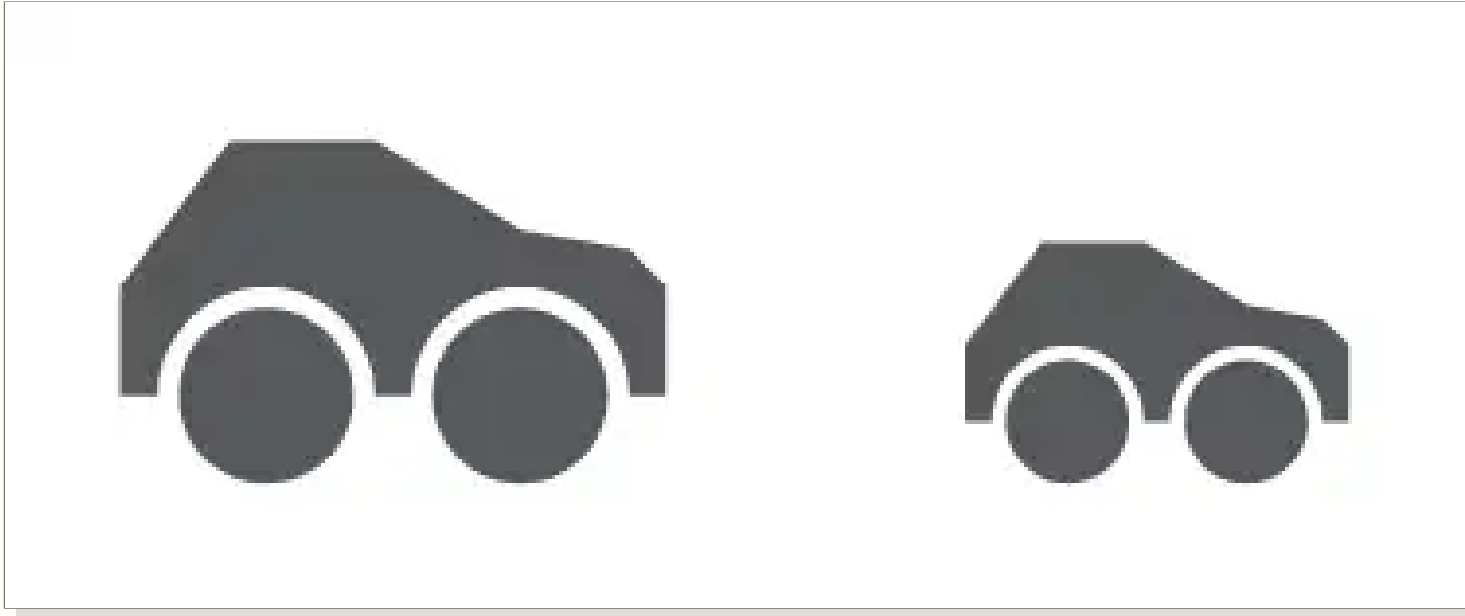
Square surfaces



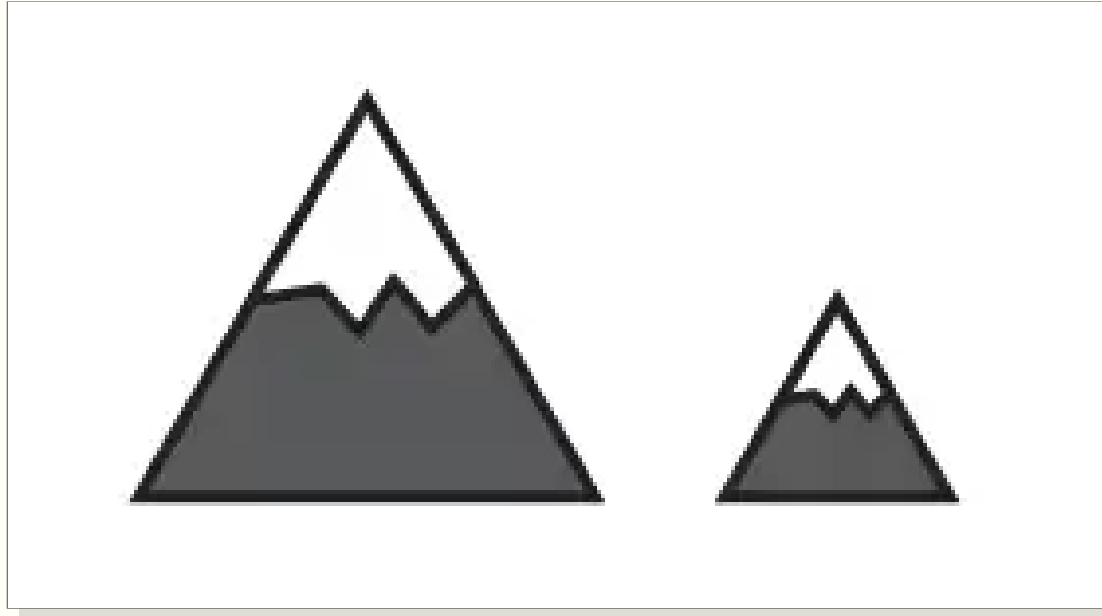
Shape surfaces



Different shape surfaces



Icon surfaces



Icon height



a



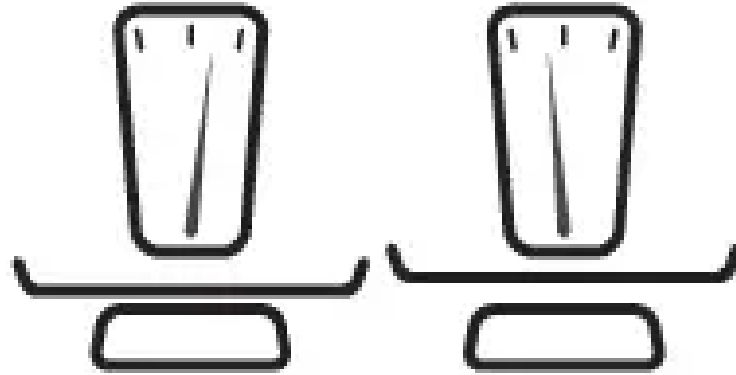
b



Volumes

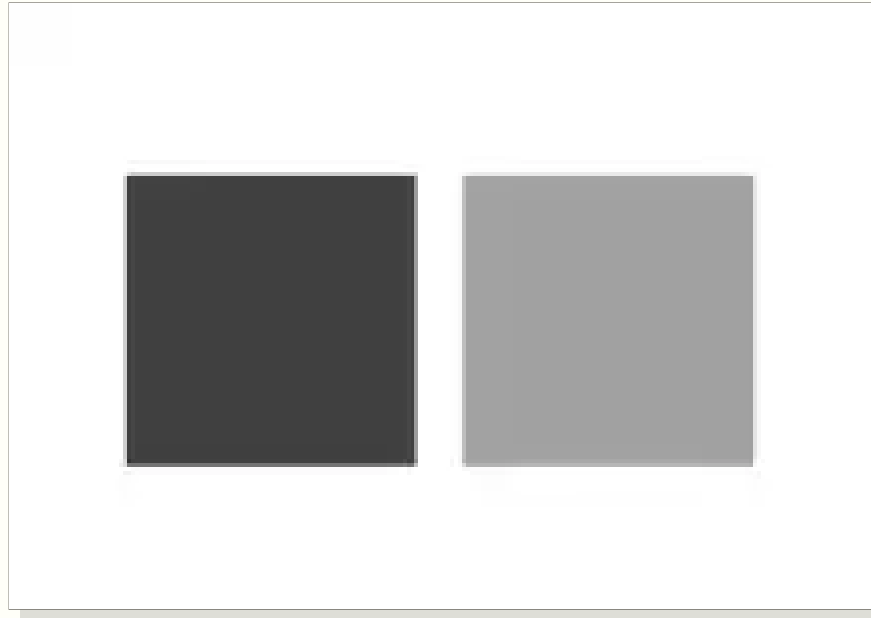


a

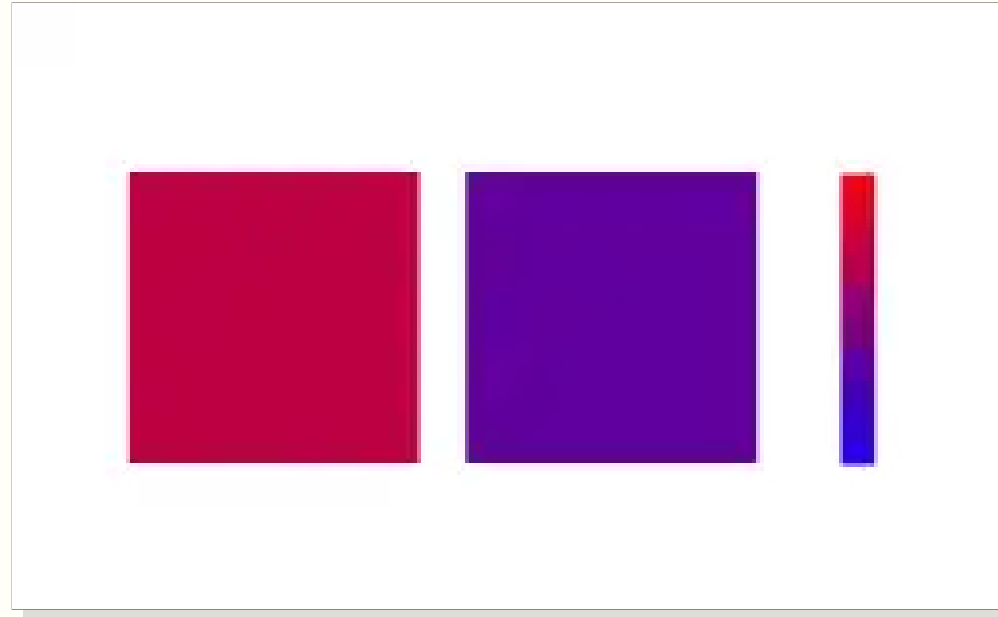


b

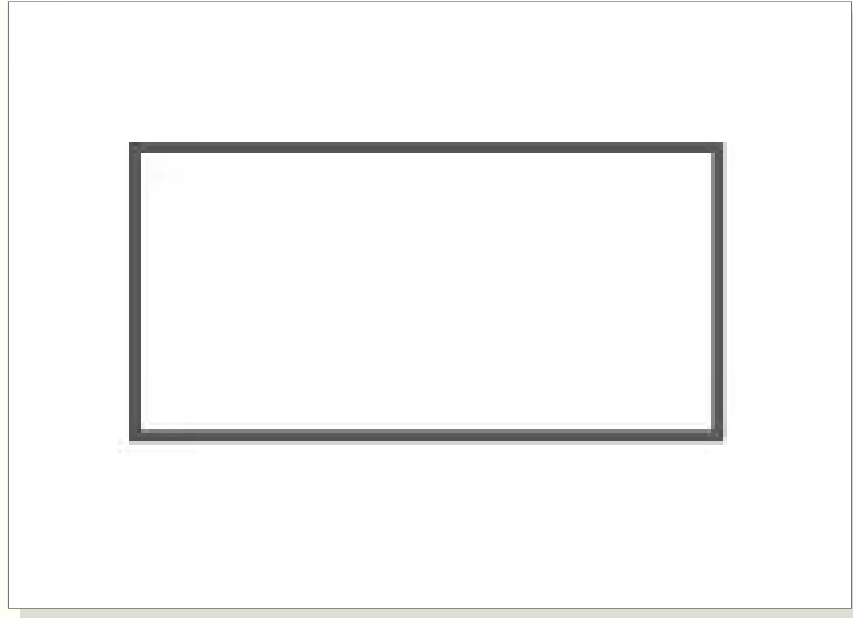
Special metaphors



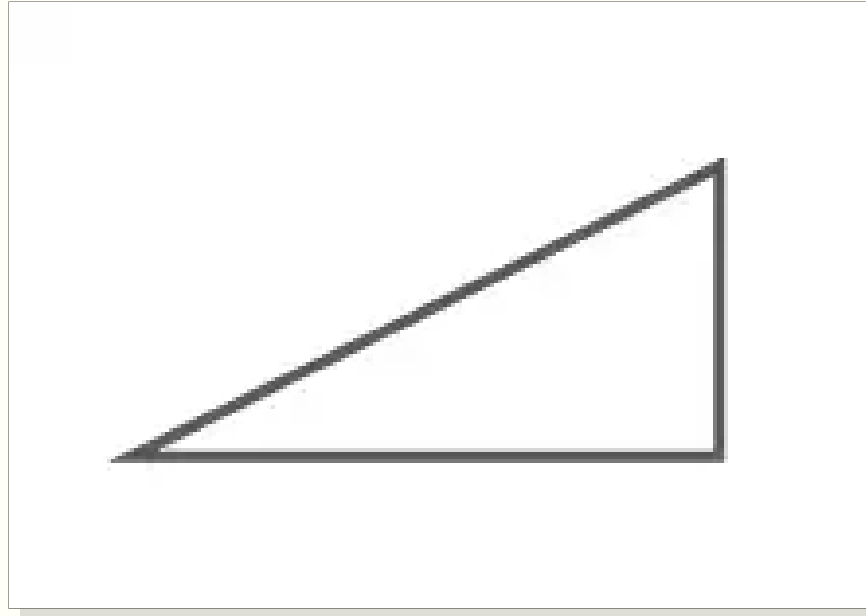
Gray tones



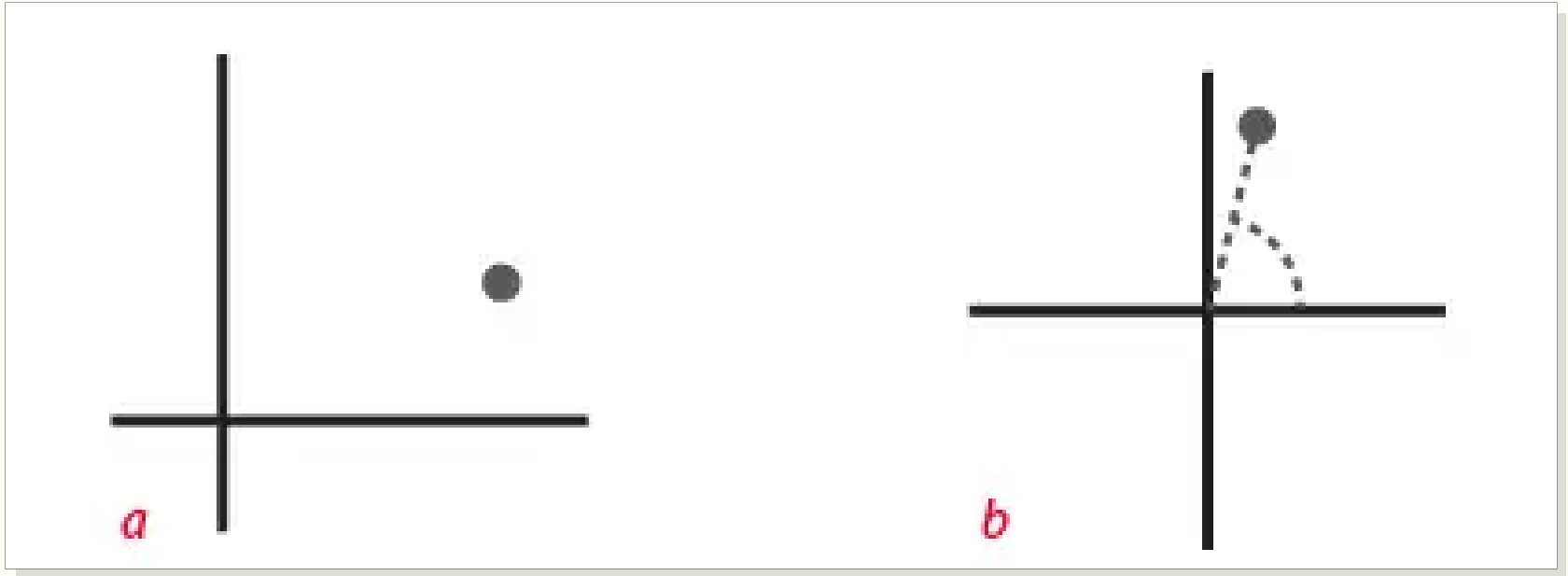
Color scale



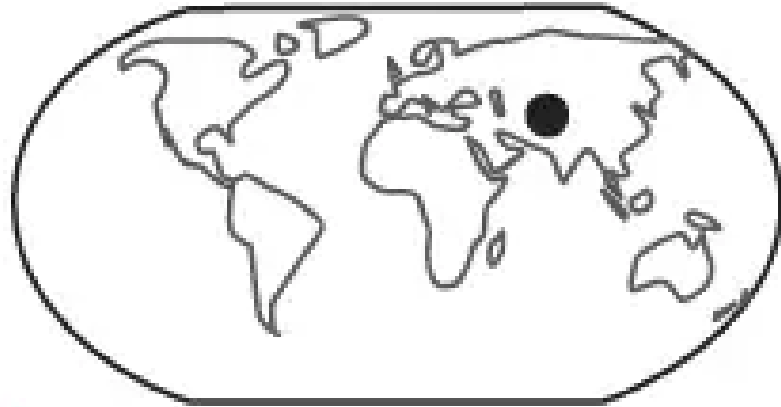
Geometric proportions



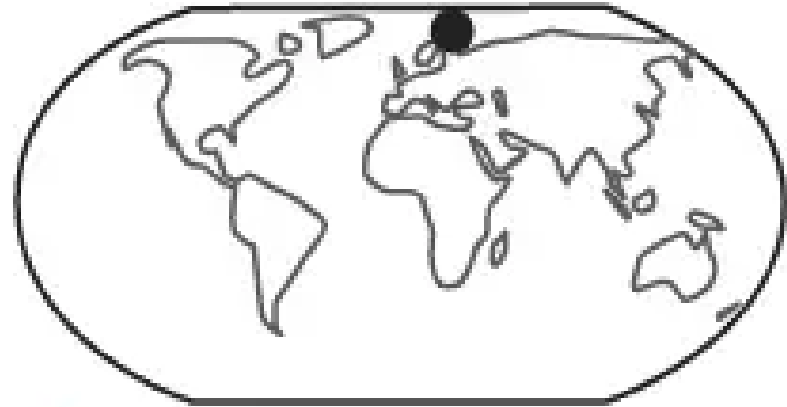
Horizontal/vertical proportions



Coordinates

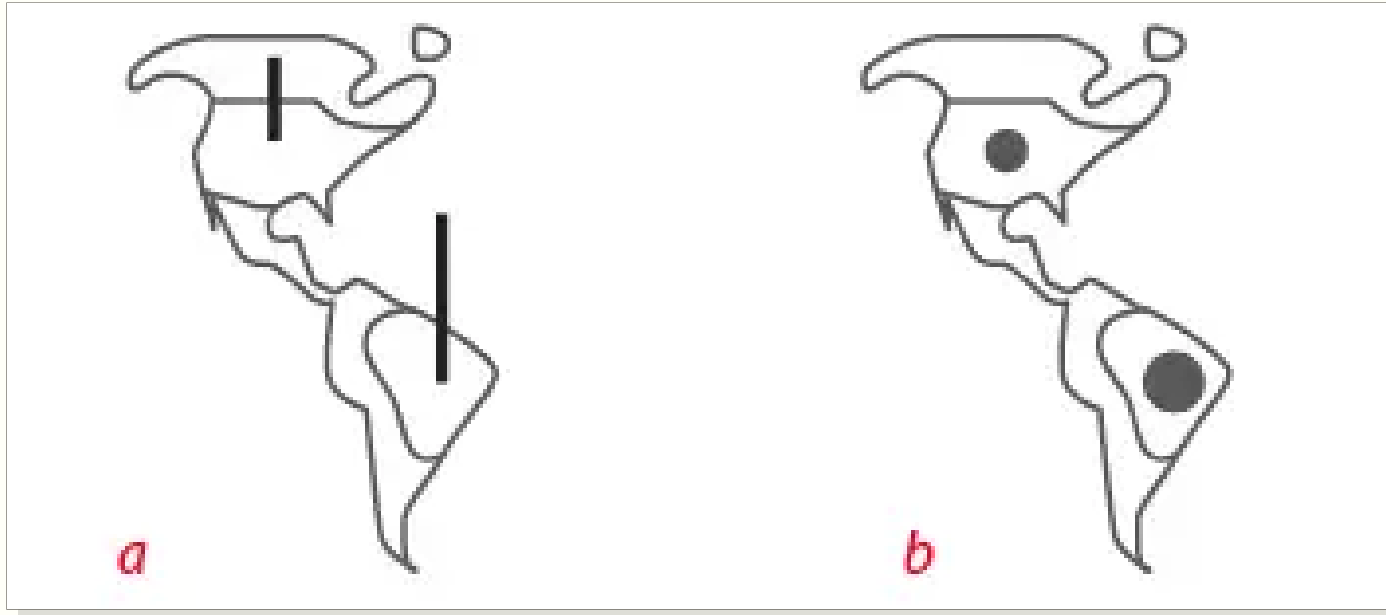


a

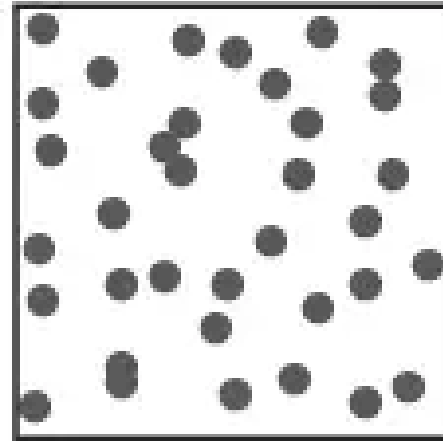
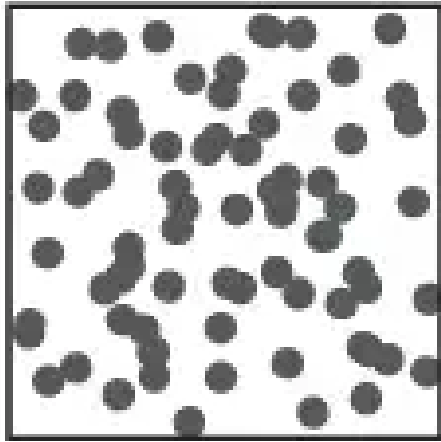


b

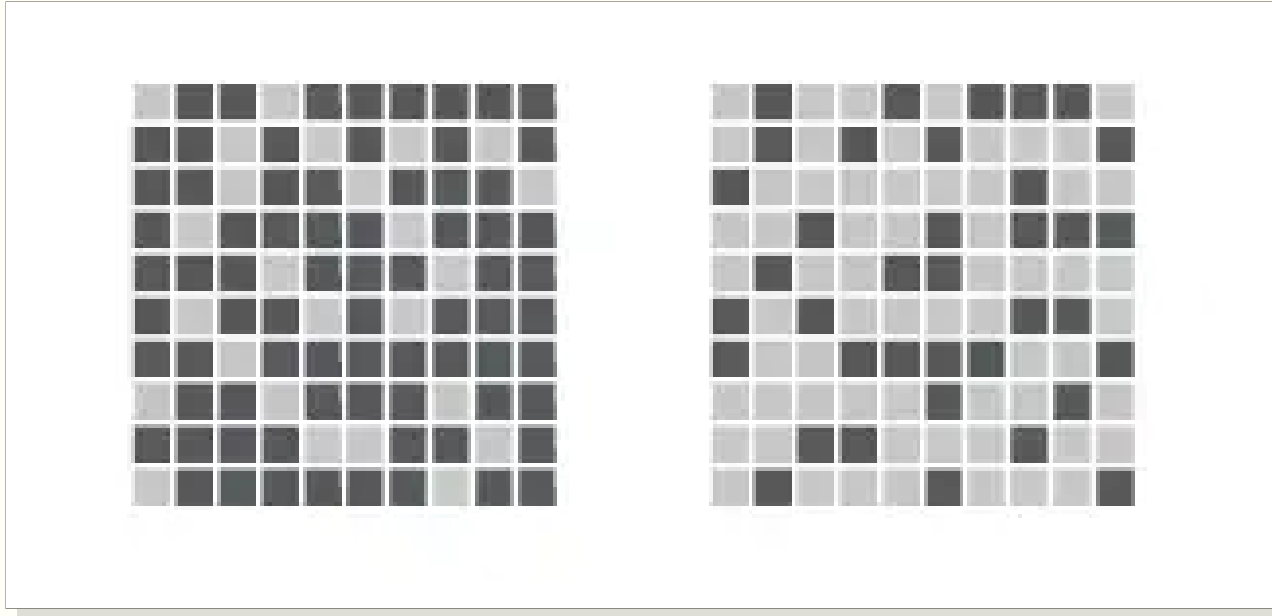
Geographic coordinates



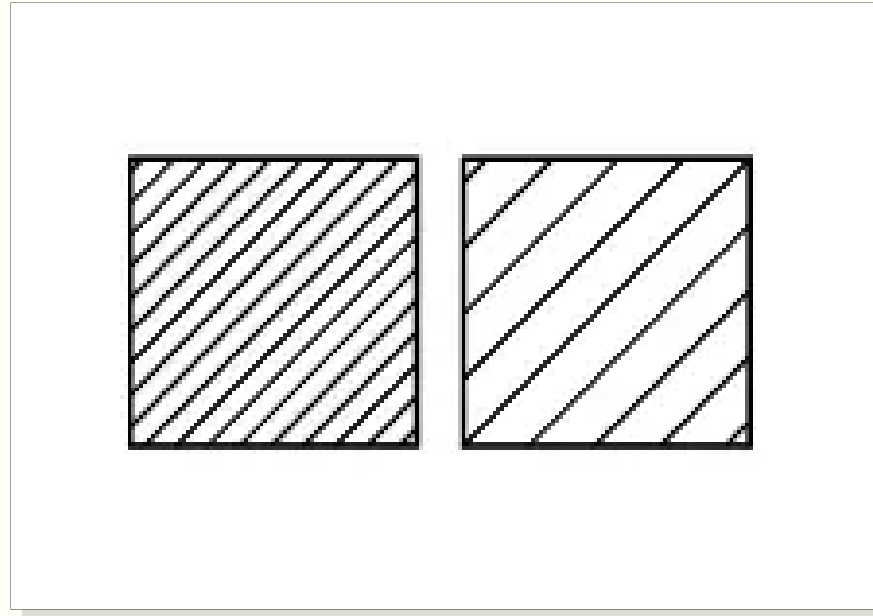
Values associated to countries



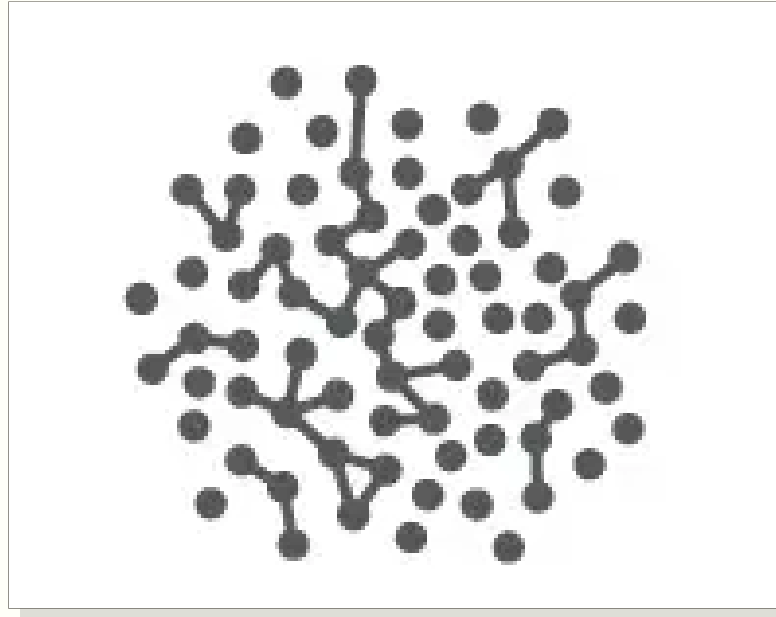
Density



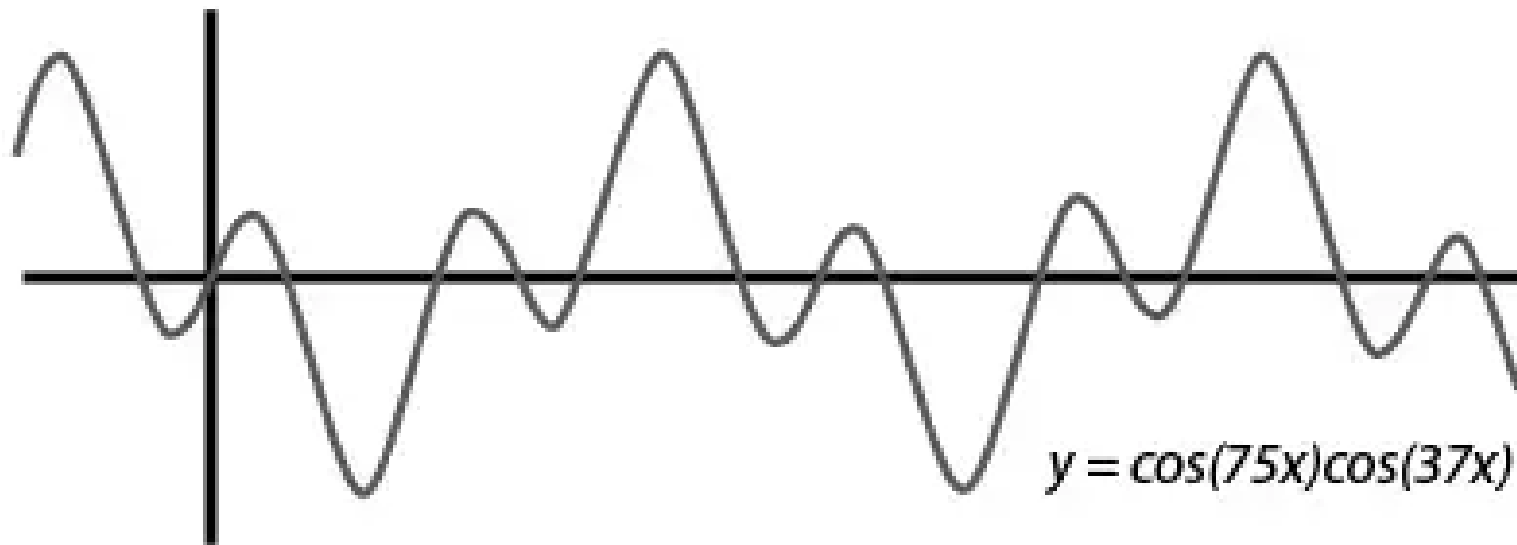
Percentages / density



Dashed

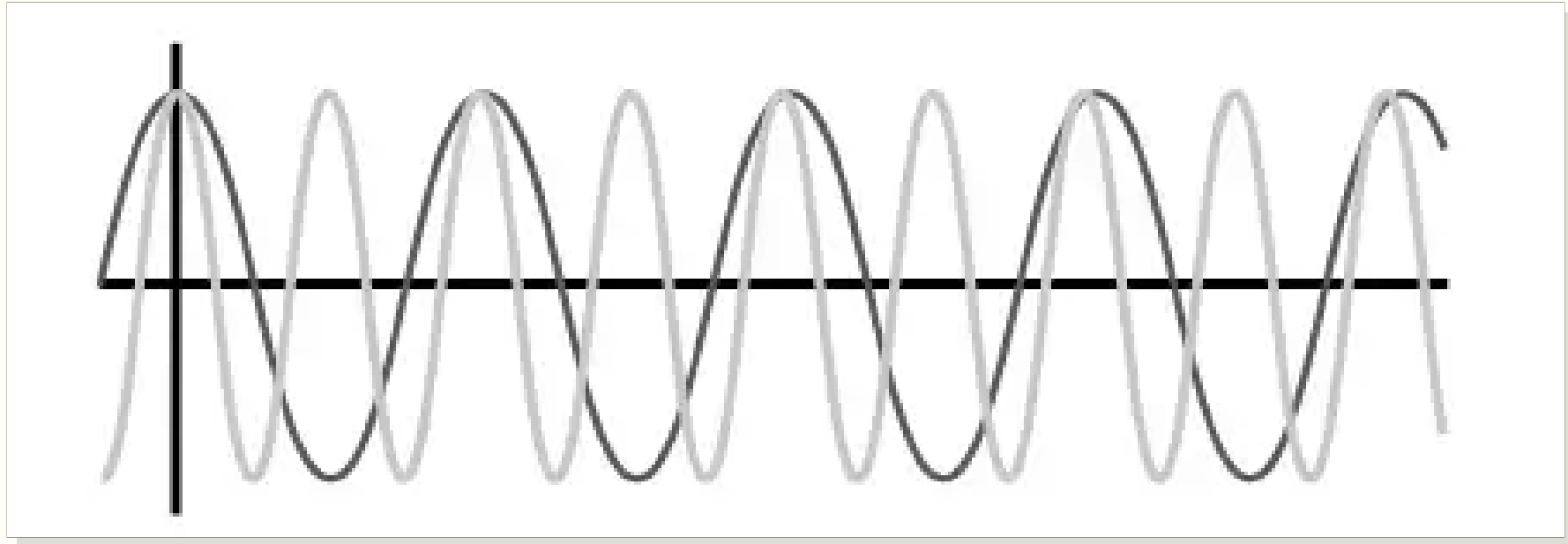


Nodes and connections

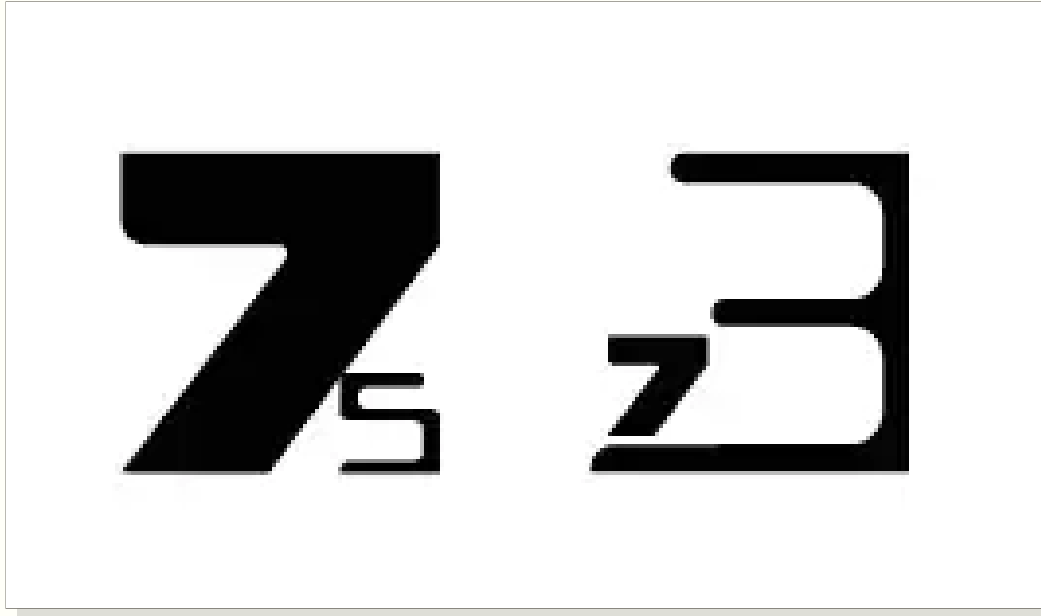


$$y = \cos(75x)\cos(37x)$$

Parameters of a function



Harmonic frequencies



Fat fonts